2026 SAIMechE Central Branch

SAIMechE
THE SOUTH AFRICAN INSTITUTION OF MECHANICAL ENGINEERING
CENTRAL BRANCH

OFFICIAL PROGRAMME

POSTGRADUATE CONFERENCE & PRIZE AWARDS

Friday, 30 October 2026

Conference: 08h30 arrival for 09h00–17h00
Gala Dinner, Panel & Awards:
17h30 arrival for 18h00–22h00

Programme Director: Ann Maria Najjuma



Programme at a glance

A structured day connecting postgraduate research, industry and academia

08h30	Registration & arrival
09h00	Opening & keynote
10h35	Student session I
12h35	Lunch
13h00	Industrial address
13h30	Student session II
16h30	Closing
17h30	Gala arrival
19h00	Dinner, panel & awards

Student presentation format 10 min presentation • 5 min Q&A • 5 min transition

Presentation protocol

A consistent format keeps the programme fair and on schedule

10 MIN

Present the abstract

Focus on problem, method, key results and engineering contribution.

5 MIN

Audience questions

Programme Director or session chair manages focused questions.

5 MIN

Presenter transition

Judges complete notes while the next presenter connects and prepares.

TOTAL PROGRAMMED TIME PER PRESENTER: 20 MINUTES

Morning programme: Opening & keynote

Programme Director: Ann Maria

08h30–09h00	Registration, arrivals and seating | Conference team
09h00–09h05	Opening and programme orientation | Ann Maria Najjuma, SaiMechE Wits Students Chairperson
09h05–09h15	Welcome address | Prof Nico Wilke, HoS - School of Mechanical, Industrial and Aeronautical Engineering
09h15–09h25	SAIMechE CEO address | Mr. Uncel Mhelembe,
09h25–09h35	SAIMechE Central Branch Chairperson’s message | John Mokwatlo
09h35–10h15	Keynote address | Dr Nkosinathi Madushela
10h15–10h35	Tea break and networking | All delegates

Keynote address: Dr Nkosinathi Madushele

Professional biography



Current role. Chief Engineer (Mechanical), Gauteng Department of Infrastructure Development, Directorate of Engineering Services.

Professional credentials. ECSA-registered Professional Engineer (PrEng), Certified Energy Manager and NRF Y2-rated researcher.

Qualifications. Doctorate in Mechanical Engineering, master's in Engineering Management and bachelor's in Mechanical Engineering, all from the University of Johannesburg.

Academic leadership. Former Associate Professor and Head of Department at the University of Johannesburg. He has supervised master's and doctoral research and mentored candidate engineers.

Research interests. Energy systems, life cycle assessment, sustainability, condition monitoring, machine learning and digital twins, connecting research with practical applications.

Engineering practice. His experience spans industry, academia and public-sector engineering. He provides technical oversight and review of mechanical infrastructure projects and develops engineers with sound judgement and professional responsibility.

Technical session I: Room 1 (1 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

10h35–10h55

01 A.C. Golozeleni, M. Lucas

A Systematic Literature Review on the Viability and Implementation Strategies of Cold Spray Additive Manufacturing for Maintenance and Reliability Improvement Initiatives on Structural Aircraft Components: A South African Air Force PC-7 MkII Case Study

10h55–11h15

02 Ann Maria Najjuma

Finite element analysis and machine learning for predicting plastic strain in eroded boiler tubes: a comparative study

11h15–11h35

03 B Emwanu, A Panday, L Ntombela

Translating e-hailing dynamics to the skies: optimal pricing strategies and vehicle class segmentation for urban air mobility

Parallel rooms. 10 min presentation, 5 min Q&A and 5 min transition per abstract.

Technical session I: Room 1 (2 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

11h35–11h55	**04 Dhruti Dheda** An engineer’s missing objective: designing energy systems for people, not just cost and carbon
11h55–12h15	**05 Duncan Goqwane** Integrated inventory and degradation modelling for optimizing flotation reagent management in a platinum concentrator
12h15–12h35	**06 G. van den Berg & D.N. Wilke** 3D object detection and instance counting in gaussian splat scenes across controlled geometric primitives
12h35–13h00	Lunch and networking

Parallel rooms. 10 min presentation, 5 min Q&A and 5 min transition per abstract.

Technical session I: Room 2 (1 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

10h35–10h55

13 Musawenkosi Nkosi

Effectiveness of a cold sprayed, selfsanitising, surface coating in reducing pathogen contamination on high-touch surfaces in the neonatal wards of Chris Hani Baragwanath Academic Hospital

10h55–11h15

14 Naveshin Jawaheer

From Print to Practice: Evaluating the Durability of 3D-Printed PLA for ColdSprayed Antimicrobial Components

11h15–11h35

15 Nokutenda Sandangwa

CFD Investigation of Winglet Cant Angle Effects on the Aerodynamic Performance of a NACA 2412 Aircraft Wing

Parallel rooms. 10 min presentation, 5 min Q&A and 5 min transition per abstract.

Technical session I: Room 2 (2 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

11h35–11h55	**16 Rachael Michael** Cfd investigation of blade tip modifications on the aerodynamic performance of the nrel phase vi wind turbine
11h55–12h15	**17 Refilw Ndlela** A systems-based comparative study of non-destructive testing (ndt) methods for thermite weld assessment
12h15–12h35	**18 Rucille Ramlal** Prediction of the torsional stiffness of a 24 kv marine power cable using a simplified finite element model
12h35–13h00	Lunch and networking

Parallel rooms. 10 min presentation, 5 min Q&A and 5 min transition per abstract.

Industrial address | Grace Olukune

Senior Engineering Manager, Renewable Energy | Eskom



BIOGRAPHY

Grace Olukune is the Senior Engineering Manager for Renewable Energy at Eskom. She is a professionally registered engineer with more than 21 years of industry experience and holds a Bachelor of Science in Electrical Engineering from Michigan State University and a Master's degree in Industrial Engineering from Wits University.

Since joining Eskom in 2007 as a Senior Engineer, Grace has held technical and leadership positions across the Generation business, including Chief Engineer and Engineering Group Manager at Kusile Power Station. She has contributed to Eskom's new-build projects from early development through to Generation Operations, providing technical leadership, multidisciplinary engineering coordination and strategic advisory support.

Grace later transitioned to Eskom's Renewables business, where she helps build the engineering capability required to support the just energy transition. Beyond her professional achievements, she is passionate about developing women and young professionals and empowering them to become impactful leaders.

RECOGNITION

- Kusile Manager of the Year
- IEEE Women in Engineering Female Engineer of the Year
- WiEBE Award for Excellence in Industry at Company Level
- Rising Star Award: Energy and Environment

Engineering leadership for a just energy transition

Afternoon programme: Room 1 (1 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

13h00–13h30	Industrial address | Grace Olukune
13h30–13h50	**07 Gary Strobel, Anthony Mutemi** An analysis of the case for unattended boilers.
13h50–14h10	**08 Harry Saner, Dr. Bevan Smith, Dr. Ponstho Twala** Autonomous and mapless navigation with reinforcement learning (rl) and transfer learning (tl) for unmanned aerial vehicles (uav) in underground connection-denied environments.
14h10–14h30	**09 J Cardoza, K D Nyembwe & Y D Mfoumboulou** Application and dimensional capabilities of voxeljet vx1000 binder jetting in metal casting: a prismaguided systematic review

Afternoon programme: Room 1 (2 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

14h30–14h50	**10 J.E. Ngobeni, M.D.I. Lucas** Establishing standardised metallographic protocols for accurate microstructural characterisation in cold spray additive manufacturing
14h50–15h10	**11 JM Kamerman & DN Wilke** Diagnosing physical fidelity in neural operators: a mode-by-mode framework for validating learned solvers
15h10–15h30	Refreshment break
15h30–15h50	**12 Martin Atshipara** A reliability-integrated risk-based maintenance strategy selection framework using promethee for heavy mining equipment: a case study of komatsu haul trucks
15h50–16h10	Reserve time
16h10–16h30	Reserve time
16h30–17h00	Closing session, acknowledgements and transition to gala

Afternoon programme: Room 2 (1 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

13h00–13h30	Industrial address | Grace Olukune
13h30–13h50	**19 Sheldon Machabe** Demand forecasting and dynamic driver allocation in on demand food delivery systems
13h50–14h10	**20 Tamzyn Alexander** Eigenstrain modelling and scan-based curvature analysis of laser shock peened aa2024-t351 almen strips
14h10–14h30	**21 W. Mokasi & D.N. Wilke** Evaluating edge-ai architectures for cold-chain monitoring: a multiobjective systems engineering approach

Afternoon programme: Room 2 (2 of 2)

Provisional allocation in PDF order. Listed authors shown. Presenting authors to be confirmed.

14h30–14h50	**22 Zamahlase Sibisi** An investigation into prolonged patient waiting times and systemic barriers to equitable public healthcare in south africa: a case for universal health coverage and advanced technology integration.
14h50–15h10	**23 Zamokuhle luthuli** Advanced ultrasonic signal feature extraction for in-process monitoring of msla 3d printing
15h10–15h30	Refreshment break
15h30–15h50	Reserve time
15h50–16h10	Reserve time
16h10–16h30	Reserve time
16h30–17h00	Closing session, acknowledgements and transition to gala

PRESENTATIONS COMPLETE

Thank you to all our presenters

Gala dinner & prize awards

An evening of celebration and connection

Arrival from 17h30
Formal programme 18h00–22h00

Gala Dinner, Panel Discussion & Prize Awards

Arrival from 17h30 | Formal programme 18h00–22h00

17h30–18h00	Guest arrival, registration and networking
18h00–18h10	Welcome and gala programme opening
18h10–19h00	Dinner service and networking
19h00–19h30	Sponsor–academia panel discussion
19h30–20h15	Prize awards and recognition
20h15–20h30	Acknowledgements and closing remarks
20h30–22h00	Dinner dance and networking

Panel focus: dialogue between sponsors and academia • Duration: 30 minutes